

C. Partitioning the Array

time limit per test: 3 seconds
memory limit per test: 256 megabytes

Allen has an array $a_1, a_2, \dots, a_n$. For every positive integer k that is a divisor of n , Allen does the following:

- He partitions the array into $\frac{n}{k}$ disjoint subarrays of length k . In other words, he partitions the array into the following subarrays:

$$[a_1, a_2, \dots, a_k], [a_{k+1}, a_{k+2}, \dots, a_{2k}], \dots, [a_{n-k+1}, a_{n-k+2}, \dots, a_n]$$

- Allen earns one point if there exists some positive integer m ($m \geq 2$) such that if he replaces every element in the array with its remainder when divided by m , then all subarrays will be identical.

Help Allen find the number of points he will earn.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the array a .

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$) — the elements of the array a .

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the number of points Allen will earn.

Example

input	Copy
8 4 1 2 1 4 3 1 2 3 5 1 1 1 1 1 6 1 3 1 1 3 1 6 6 2 6 2 2 2 6 2 6 3 6 6 6 10 1 7 5 1 4 3 1 3 1 4 1 1	
output	Copy
2 1 2 4	

Codeforces Round 919 (Div. 2)

Finished

Practice



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Language: PyPy 3.10 (7.3.15, 64bit)

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brute force math number theory *1600

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```
4
1
2
1
```

Note

In the first test case, $k = 2$ earns a point since Allen can pick $m = 2$ and both subarrays will be equal to $[1, 0]$. $k = 4$ also earns a point, since no matter what m Allen chooses, there will be only one subarray and thus all subarrays are equal.

In the second test case, Allen earns 1 point for $k = 3$, where his choice for m does not matter.

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